(*b*) Assume that the size of an organic type software product has been estimated to be 32000 lines of source code. Determine the effort required to develop software product and the nominal development time. 8

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(*c*) Schematically draw the architecture of a CASE environment and explain how the different tools are integrated. 8

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322614(22)

BE (6th Semester)
Examination, April-May, 2016

(Old Scheme)

Software Engineering

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Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 28

Note : Attempt all questions. Each questions has three parts carrying 8 marks. Attempt any two parts from (*a*), (*b*) and (*c*).

1. (*a*) Put YES or NO in blank boxes of the following table. 8

Characteristics	Waterfall	Prototype	Spiral	RAD
Are the requirement easily defined and/or well-known?				
Is the system expected to be modified, perhaps in ways not anticipated, post development?				
Is early functionality a requirement?				
Are the user representative experts in the problem domain?				

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(Turn Over)

(*b*) Differentiate between "the model which is simplest one and developed by Royce" and "the model in which risk analysis is done". 8

(*c*) Describe System and System engineering hierarchy in detail. 8

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2. (*a*) What are the main activities carried out during requirements analysis and specification phase? What is the final outcome of these phases? 8

(*b*) In software engineering, there is a process to create and maintain a system requirement document. The overall process includes the following activities :

(*i*) Feasibility study

(*ii*) Elicitation and analysis

(*iii*) Specification

(*iv*) Validation

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Describe above activities in detail. 8

(*c*) Draw DFD for library management system which is divided into four parts : 8

(*i*) Book details

(*ii*) Issue details

(*iii*) Return details

(*iv*) Student details

3. (*a*) Differentiate betwen cohesion and coupling. 8

(*b*) During the software development life cycle or after its release software is changed several times. As a result of those changes multiple versions of software system exist at one time. SCM helps to manage and control those changes. Describe various reasons for those changes. Also describe SCM tasks. 8

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(*c*) Write short notes on :

(*i*) Data design

(*ii*) Architectural design

(*iii*) Interface design

(*iv*) Component level design 8

4. (*a*) Differentiate between Glass box testing and Behavioral testing. 8

(*b*) How is cyclomatic complexity calculated in a flow graph? Describe with an appropriate example. 8

(*c*) Describe the different approaches of integration testing. 8

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5. (*a*) Suppose you are the project manager of a large product development team and you have to make choice between democratic and chief programmer team organizations. Which one you adopt and why? 8